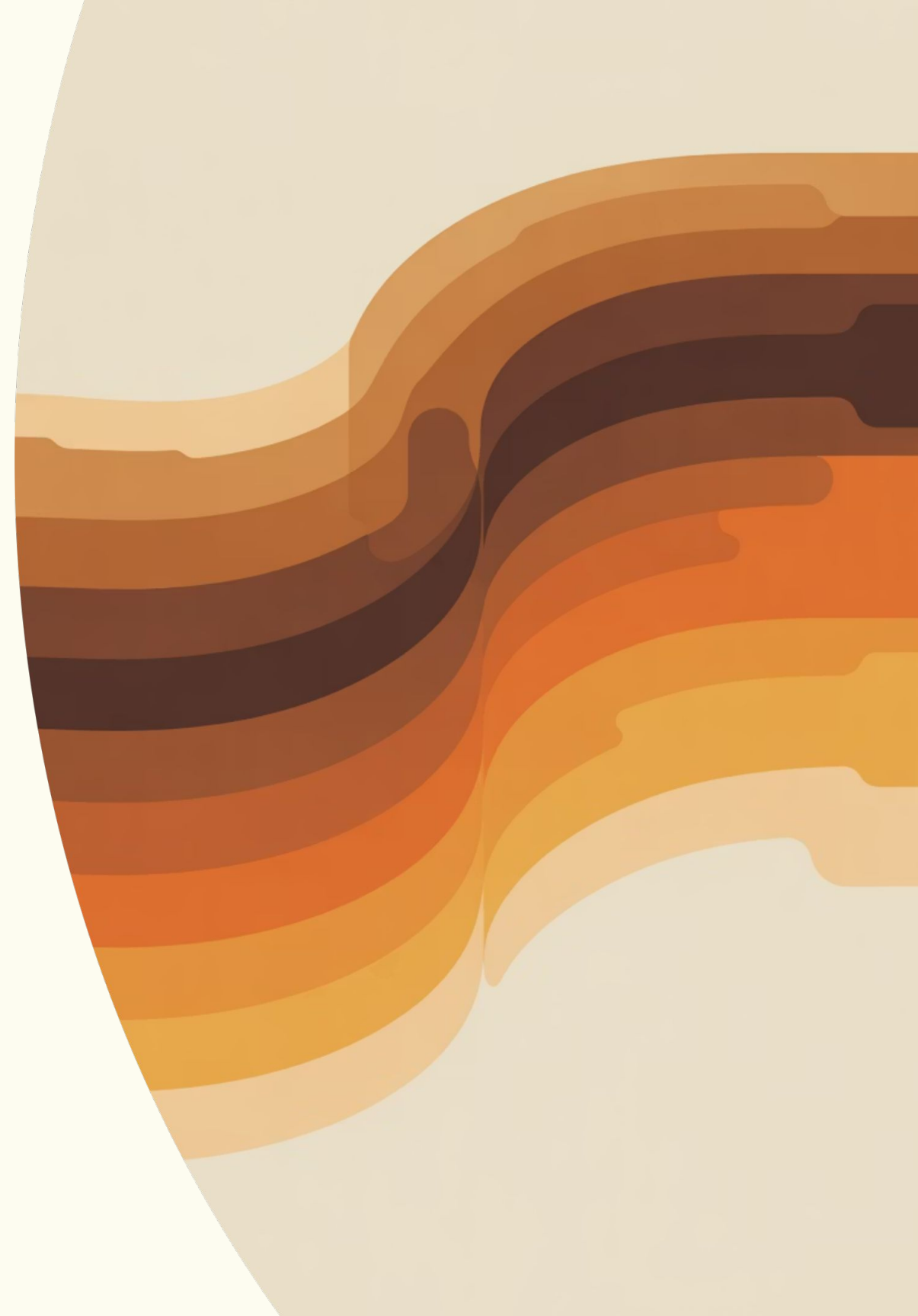


# ROC Curve, AUC, and Threshold Interpretation

Module 11.6 — Understanding probability thresholds and model performance evaluation in healthcare machine learning



# Why Probability Matters in Clinical Decisions

1

Decision trees and most ML models don't output simple **Yes/No** predictions. Instead, they generate **probability scores** that reflect confidence levels.

2

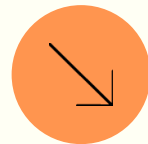
These probabilities provide nuanced information about prediction certainty, enabling more informed clinical decision-making.



**High Confidence**

**0.82** probability

Strong indication of disease presence



**Low Confidence**

**0.30** probability

Lower likelihood of disease



**Decision Point**

**0.50** default threshold

Where predictions split

# What Is a Classification Threshold?

A **threshold** is the **decision boundary** that determines when to label a prediction as positive. It's the **cutoff point** where probability converts to action.

1

## Threshold Rule

If probability  $\geq 0.5 \rightarrow$  Predict "Heart Disease"

2

## Impact on Metrics

Changing threshold adjusts TP, FP, FN, precision, and recall

3

## Clinical Tuning

Optimal threshold depends on medical context and risk tolerance

# Why Threshold Selection Matters in Healthcare



## High Threshold (0.7+)

- Fewer false alarms
- More missed patients
- Higher precision, lower recall

## Low Threshold (0.3)

- Fewer missed patients
- More false alarms
- Lower precision, higher recall

Medical scenarios require **careful threshold tuning** based on clinical consequences, not blind acceptance of the 0.5 default.

# The ROC Curve: Visualizing Model Performance

The **Receiver Operating Characteristic (ROC)** curve is a powerful visualization tool that evaluates classification performance across *all possible thresholds* simultaneously.

O1

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## True Positive Rate

Proportion of actual positives correctly identified (Recall)

O2

---

## False Positive Rate

Proportion of actual negatives incorrectly classified as positive

O3

---

## Threshold Points

Each point on the curve represents one threshold value

O4

---

## Class Separation

ROC reveals how well the model distinguishes between classes

# Understanding ROC Curve Axes



## X-Axis: False Positive Rate

$$\text{FPR} = \text{FP} / (\text{FP} + \text{TN})$$

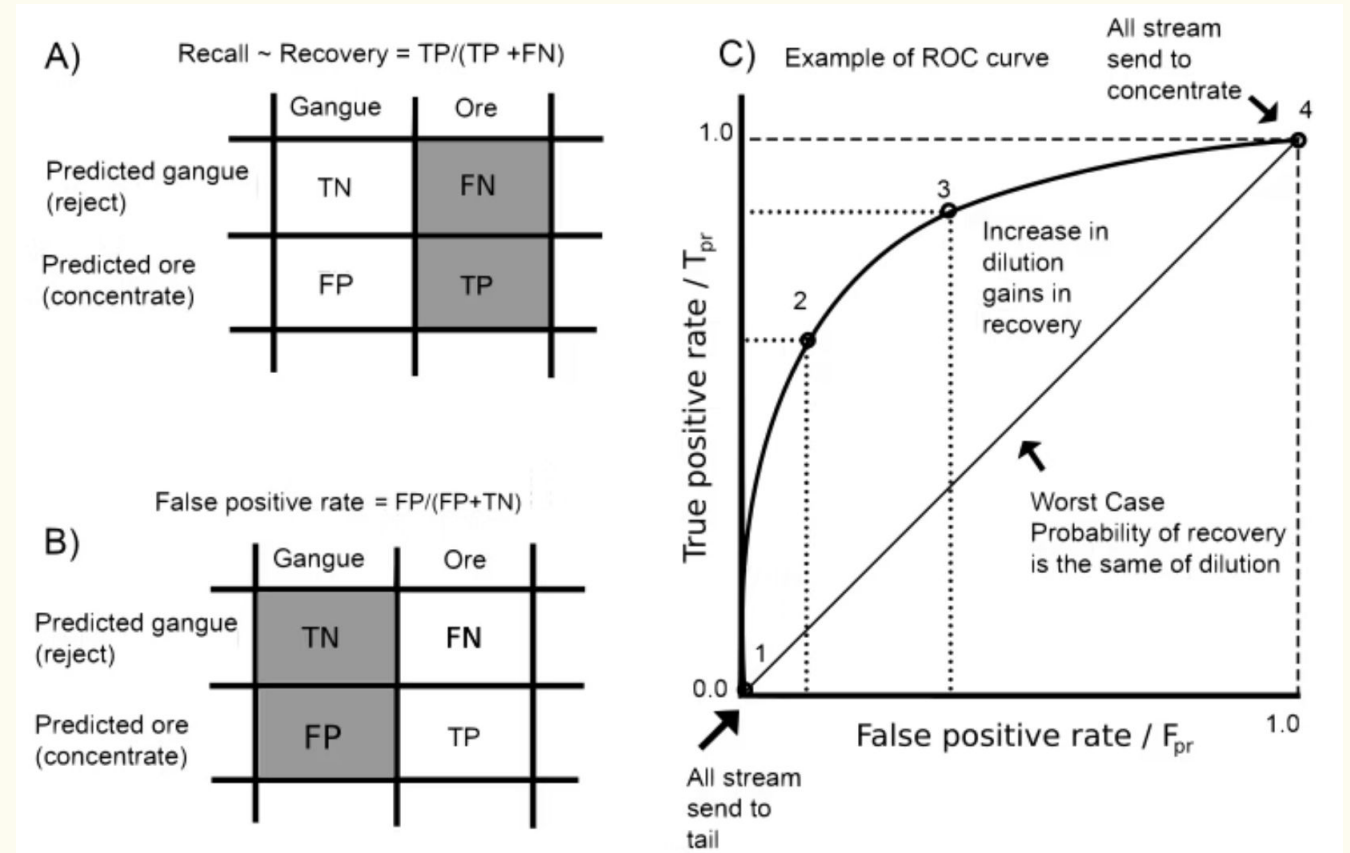
Measures the proportion of healthy patients incorrectly flagged as having disease



## Y-Axis: True Positive Rate

$$\text{TPR} = \text{TP} / (\text{TP} + \text{FN})$$

Also called Recall — measures the proportion of sick patients correctly identified



The ideal model achieves **high TPR** (catching most disease cases) **while maintaining low FPR** (minimizing false alarms).

# Interpreting ROC Curve Shape



## Good Model

Smooth curve bending toward  
top-left corner



## Random Guessing

Diagonal line from bottom-left to  
top-right



## Broken Model

Curve falling below the diagonal  
— predictions are inverted

The closer the curve hugs the top-left corner, the better the model's discriminative ability across all threshold settings.

# AUC: Summarizing ROC Performance

**Area Under the Curve (AUC)** condenses the entire ROC curve into a single interpretable metric, representing the probability that the model ranks a random positive case higher than a random negative case.

1.0

Perfect

Flawless separation

0.9

Excellent

Strong performance

0.7

Acceptable

Useful for screening

0.5

Random

No better than chance

# Module 11 Complete: Decision Trees Mastery

**Tree Fundamentals**  
Structure, splits, and decision-making process

**Advanced Metrics**  
ROC curves, AUC, and threshold tuning

**Evaluation**  
Confusion matrix, precision, recall, F1-score



**Core Concepts**

Entropy, Gini impurity, information gain calculations

**Model Building**

Building and training your first tree classifier

**Optimization**

Overfitting prevention and pruning strategies

## Decision Trees Excel At:

- Interpretability for clinical stakeholders
- Quick baseline model development
- Medical explainability requirements
- Handling mixed data types

## Coming Next

Future modules will explore **ensemble methods** like Random Forests and Gradient Boosting that combine multiple decision trees for superior performance.